

# MTH210

Examination - January Semester 2026

## Fundamentals of Probability

Thursday, 16 April 2026

04:00 pm - 06:00 pm

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**Time allowed: 2 hours**

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### INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **FOUR (4)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if **FULL** working is shown.
4. This is an **Open-book** examination.

### At the end of the examination.

Please ensure that you have written your examination number on each answer book used.

### Failure to do so will mean that your work cannot be identified.

If you have used more than one answer book, please tie them together with the string provided.

**THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR  
SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.**

Answer ALL questions. (Total: 100 marks)

### Question 1

- (a) In a dice game between Alice and Brenda, two fair 6-sided dice are rolled. The rules are as follows:

- The two dice rolls are labelled  $(x, y)$ .
- If either die shows a 1 or a 2, Alice loses \$1 and Brenda wins \$1.
- However, if both dice show the same number (a “pair”), Alice wins \$1 instead. For example, (1,1), (2,2), (3,3), etc.
- Otherwise, Alice wins \$1. (This occurs when both dice do not contain 1 or 2)

- (i) Show carefully that under these original rules, the game is fair, i.e. Alice and Brenda have equal chances of winning.

(8 marks)

- (ii) Now suppose the rules change as follows:

- Pairs are no longer an exception. Every time either die shows a 1 or 2, Alice loses \$1 and Brenda wins \$1, even if it is a pair.
- Otherwise, Alice wins \$1.

Let the random variable  $W$  denote Brenda’s winnings on a single play of the modified game.

Calculate the probability that Brenda wins under the modified rules.

(6 marks)

- (iii) Determine the expected value  $E(W)$  and variance  $Var(W)$  of Brenda’s winnings under the modified rules.

(8 marks)

- (b) In a health study of 1,000 people:

- 300 people are elderly (aged 65 and above).
- 700 people are younger than 65.
- Among the elderly, 60 have hypertension.
- Among the younger people, 35 have hypertension.

Let  $H$  be the event that a randomly chosen person has hypertension, and  $O$  be the event that the person is elderly.

- (i) Compute  $P(H|O)$  and  $P(H|O^c)$  and comment on the implications of age and the risk of hypertension.

(7 marks)

- (ii) Find the overall probability  $P(H)$  that a randomly selected person has hypertension and thus determine mathematically to identify whether events  $H$  and  $O$  are independent.
- (6 marks)

### Question 2

Let  $X$  be the number of bad apples in a box of 50 apples, and  $Y$  be the number of bad oranges in a box of  $n$  oranges.

The probability that any single apple is bad is 4%, and the probability that any single orange is bad is 5%. Assume that the condition of each fruit is independent of the others.

- (a) Applying a suitable distribution, determine the mean and variance of  $X$ , the number of bad apples.
- (4 marks)
- (b) Given that the probability of finding zero bad oranges in the box is 2.75837%, determine the number of oranges ( $n$ ) in the box.
- (8 marks)
- (c) Justify whether the distributions of  $X$  (number of bad apples) and  $Y$  (number of bad oranges) can be approximated by the Poisson distribution.
- (5 marks)
- (d) Applying the Poisson distribution approximation or otherwise, calculate the probability that there are at least 8 spoiled fruits in total (i.e. bad apples plus bad oranges) in the combined boxes.
- (8 marks)

### Question 3

A factory produces steel rods whose lengths follow a normal distribution with a mean of  $\mu = 50$  cm and a standard deviation of  $\sigma = 2$  cm.

- (a) Calculate the probability that a randomly selected steel rod is shorter than 47 cm.
- (6 marks)
- (b) Find the length which marks the 90th percentile of the steel rod lengths.
- (6 marks)

- (c) A customer requires rods to be between 48 cm and 52 cm long. The factory wants at least 98% of rods to meet this specification.

Solve for the maximum standard deviation the production process can have (keeping the mean at 50 cm) so that at least 98% of rods fall between 48 cm and 52 cm.

(8 marks)

#### Question 4

- (a) A continuous random variable  $X$  has PDF:

$$f(x) = \begin{cases} \frac{3}{8} x^2, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Verify that  $f(x)$  is a valid PDF. Determine  $Var(X)$ .

(10 marks)

- (b) A factory produces electronic components. Each component has a probability of 0.1 to be defective, independently of the others.

- (i) Let  $X$  be the number of components tested until the first defect is found. By applying the appropriate model, determine the mean and variance of  $X$ .

(5 marks)

- (ii) Let  $Y$  be the number of components tested until the 3rd defect was found. By applying the appropriate model, calculate the probability that the 3rd defect was found on the 10th component tested  $P(Y = 10)$ .

(5 marks)

----- END OF PAPER -----